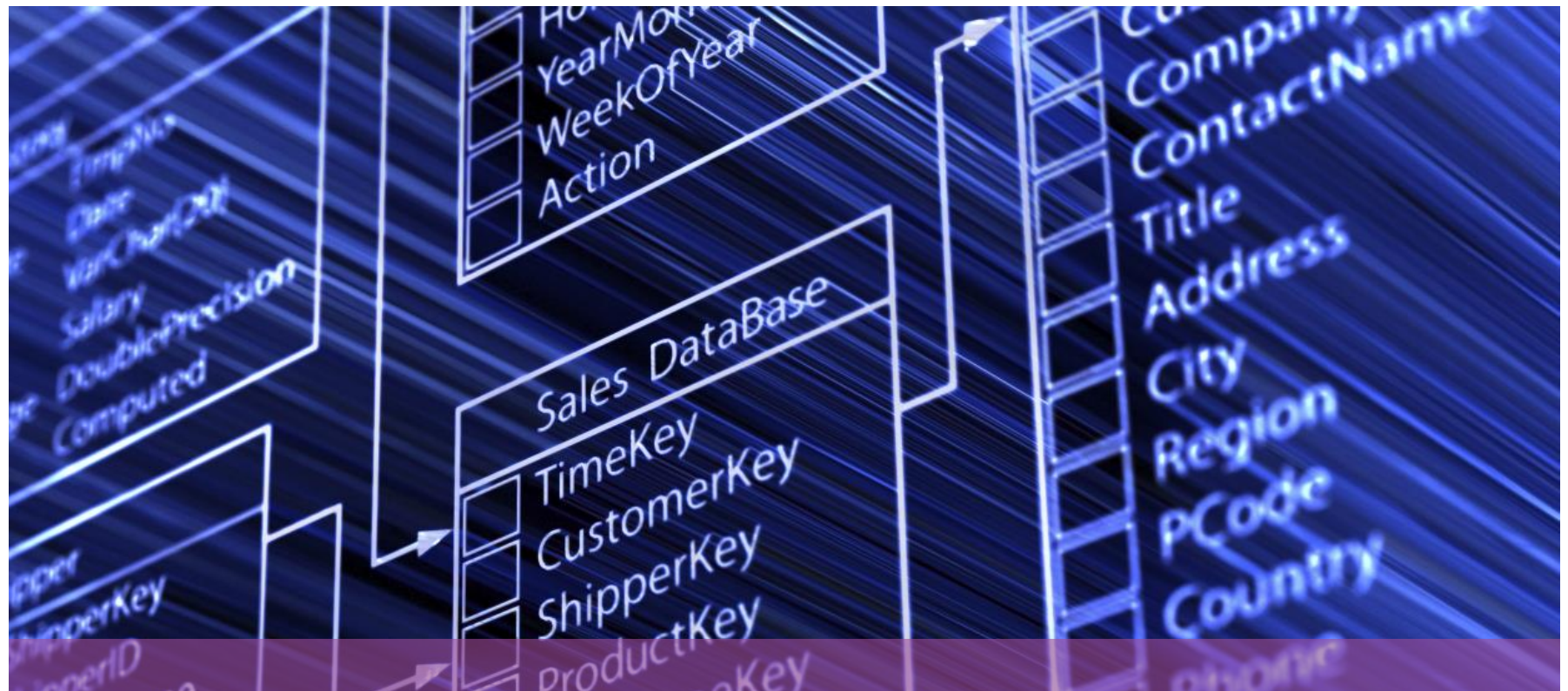




COC - Oracle Database Administration Fundamentals for SAP

Course Objectives

- Overview of database data storage model and file architecture
- Archiving process
- Using Database Administration Tools – brtools, sql plus
- Space Management
- Administrating Oracle Instances
- Introduction to SQL
- SAP DB Monitors basics
- Reorganization of segments
- Backup restore scenario
- Troubleshooting of logs files

Overview of Database

UNIT CONTENTS

- Database Architecture
- Environment Variables
- Connecting to the database

Overview of Database

Database Architecture – Basic Terms

Instance

- Active part of a database server
- It's a combination of Oracle (background) processes and memory buffers
- Every running Oracle database is associated with an Oracle instance

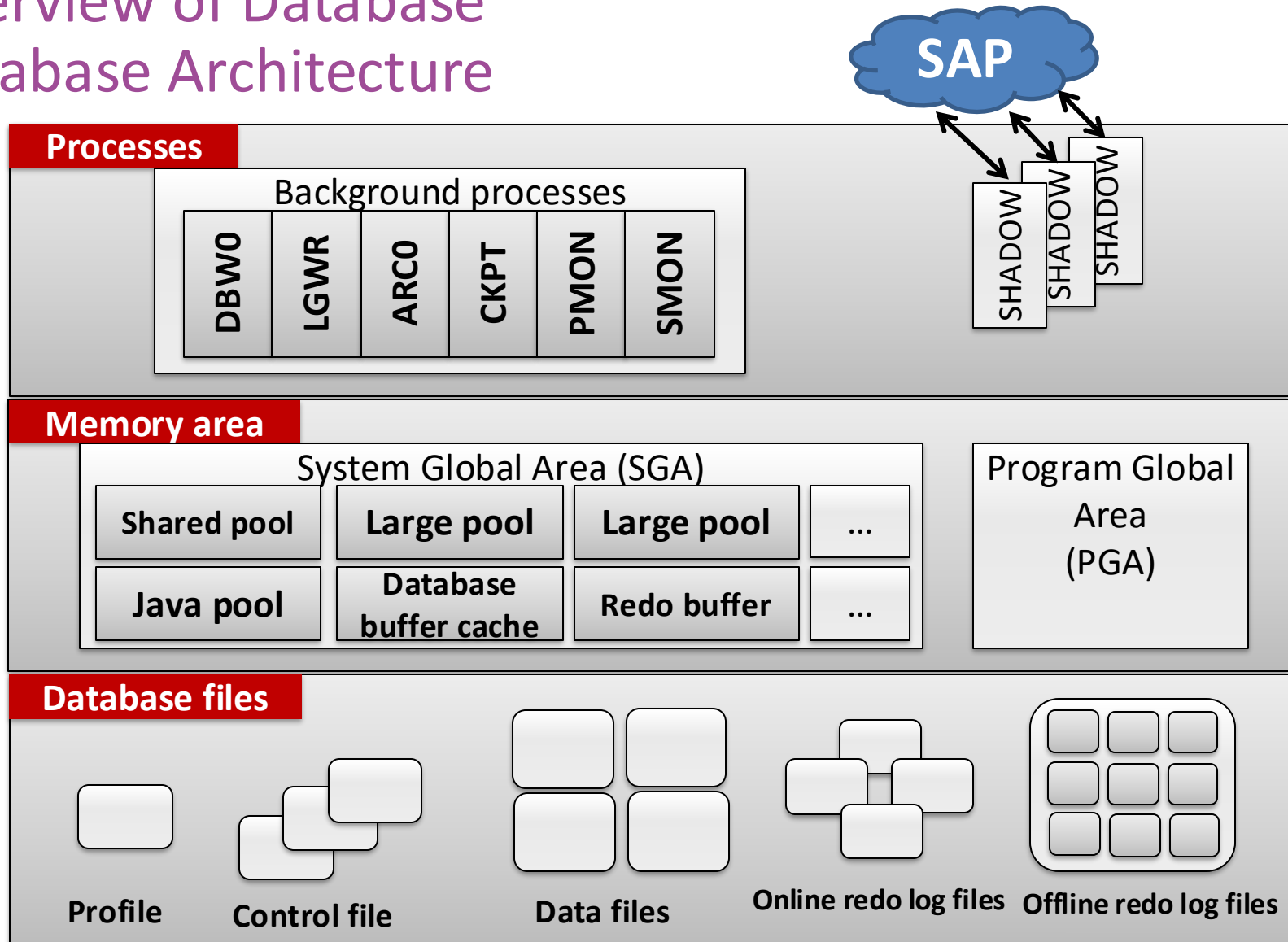
Overview of Database

Database Architecture – Basic Terms

Database

- Passive part of a database server
- It's a collection of data, logically treated as a unit
- It consists of 3 file types (data files, control files, redo log files)
- Physically, the data is stored in one or more data files on disks.
- Oracle manages database data in logical units called tablespaces. A tablespace consists of one or more data files.

Overview of Database Architecture



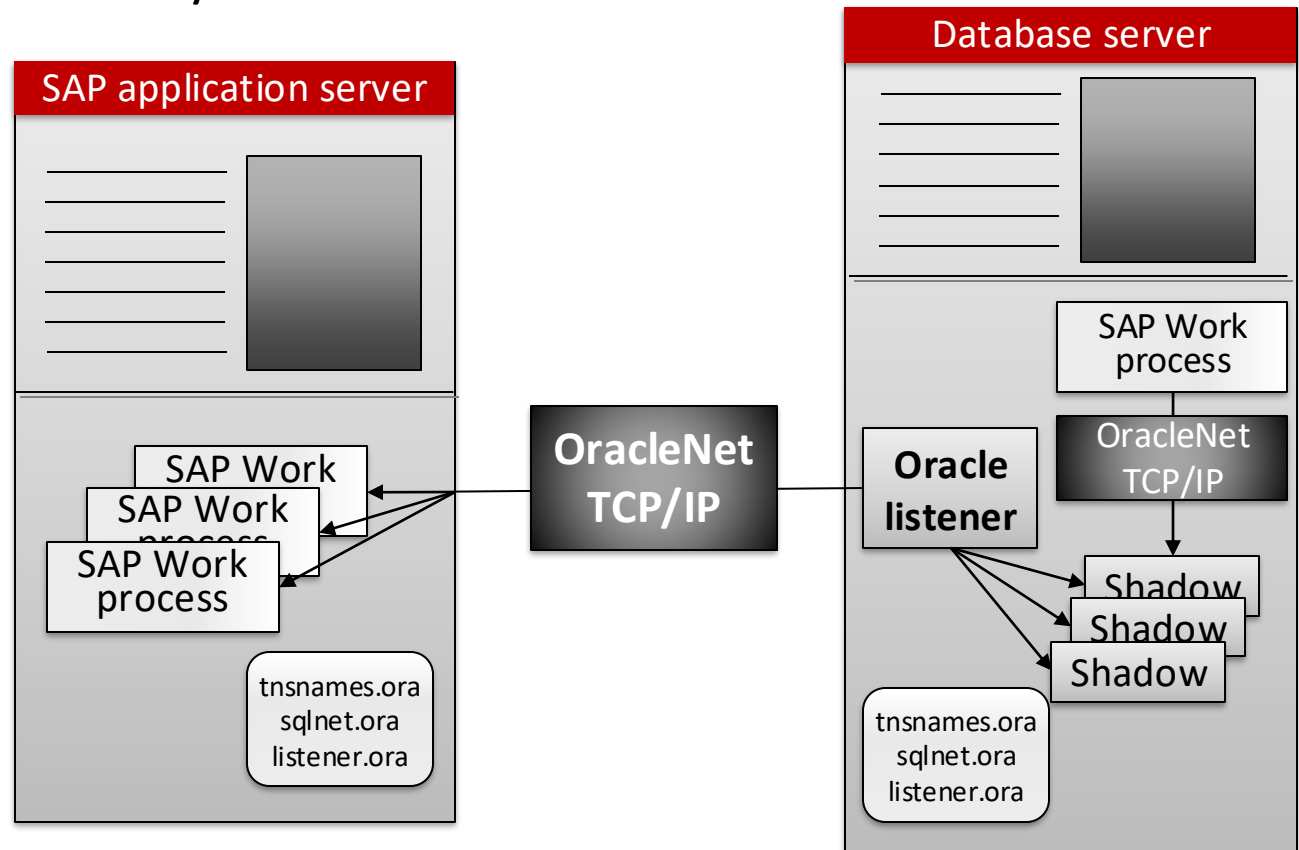
Overview of Database Listener

- Its responsibility is to listen for incoming connection requests and it forwards the requests to the server
- Once a connection is established, the client and Oracle database server communicate directly with one another
- The listener is configured with a protocol address; only clients configured with the same protocol address can send connection requests to listener
- For checking a connection to the listener, there is a tool: tnsping

Overview of Database Listener

\$ORACLE_HOME/network/admin

- listener.ora
- tnsnames.ora
- sqlnet.ora



Overview of Database Listener - managing

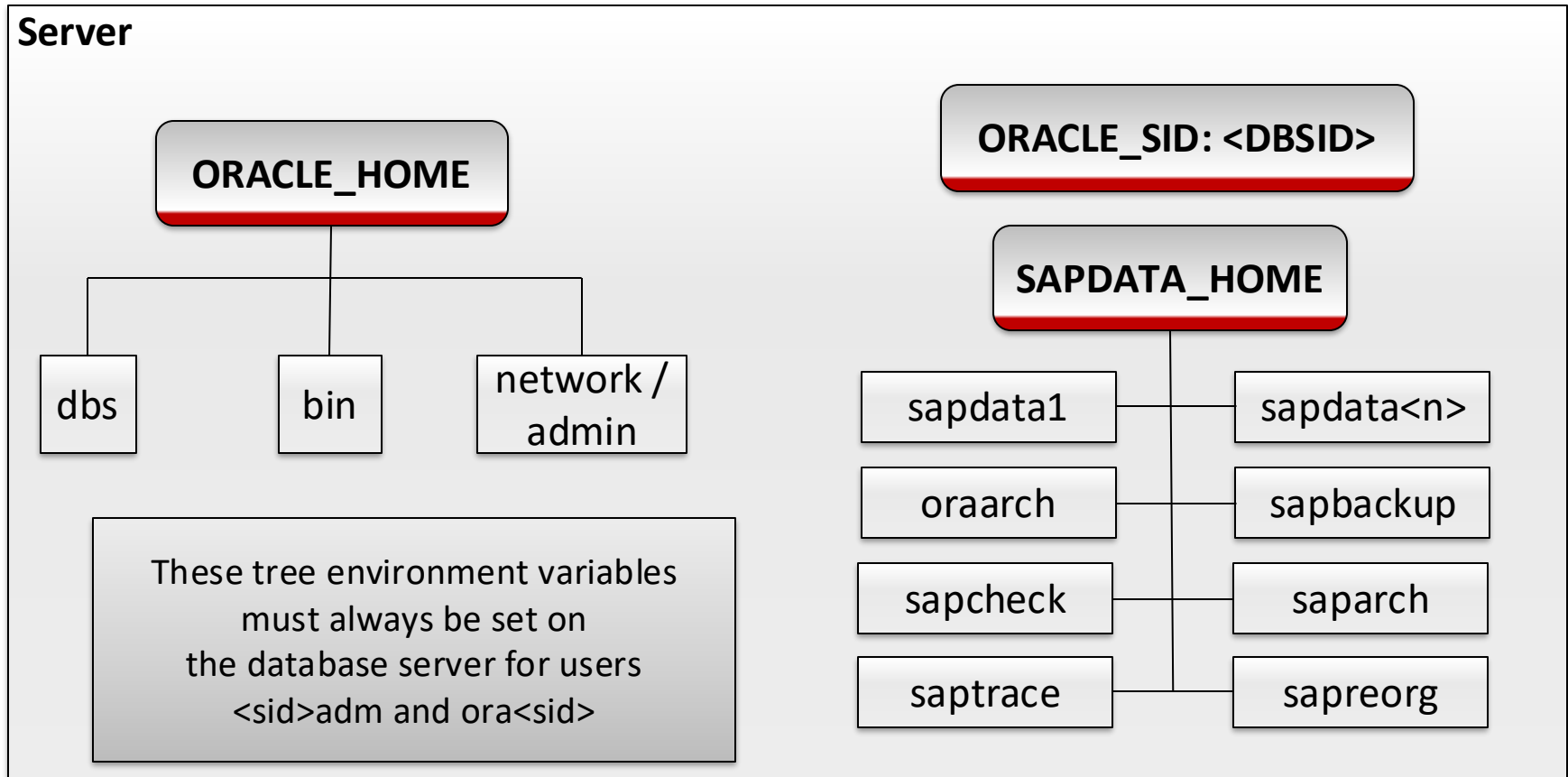
LSNRCTL> help

The following operations are available

An asterisk (*) denotes a modifier or extended command:

start	stop	status
services	version	reload
save_config	trace	spawn
change_password	quit	exit
set*	show*	

Overview of Database Oracle Directories and Environment Variables



Overview of Database Oracle Directories and Environment Variables

Displays environment variables

- `env`

Sets an environment variable to requested value (*sh and ksh shells*)

- `export <env_variable>=<value>`

or (*csH shell*)

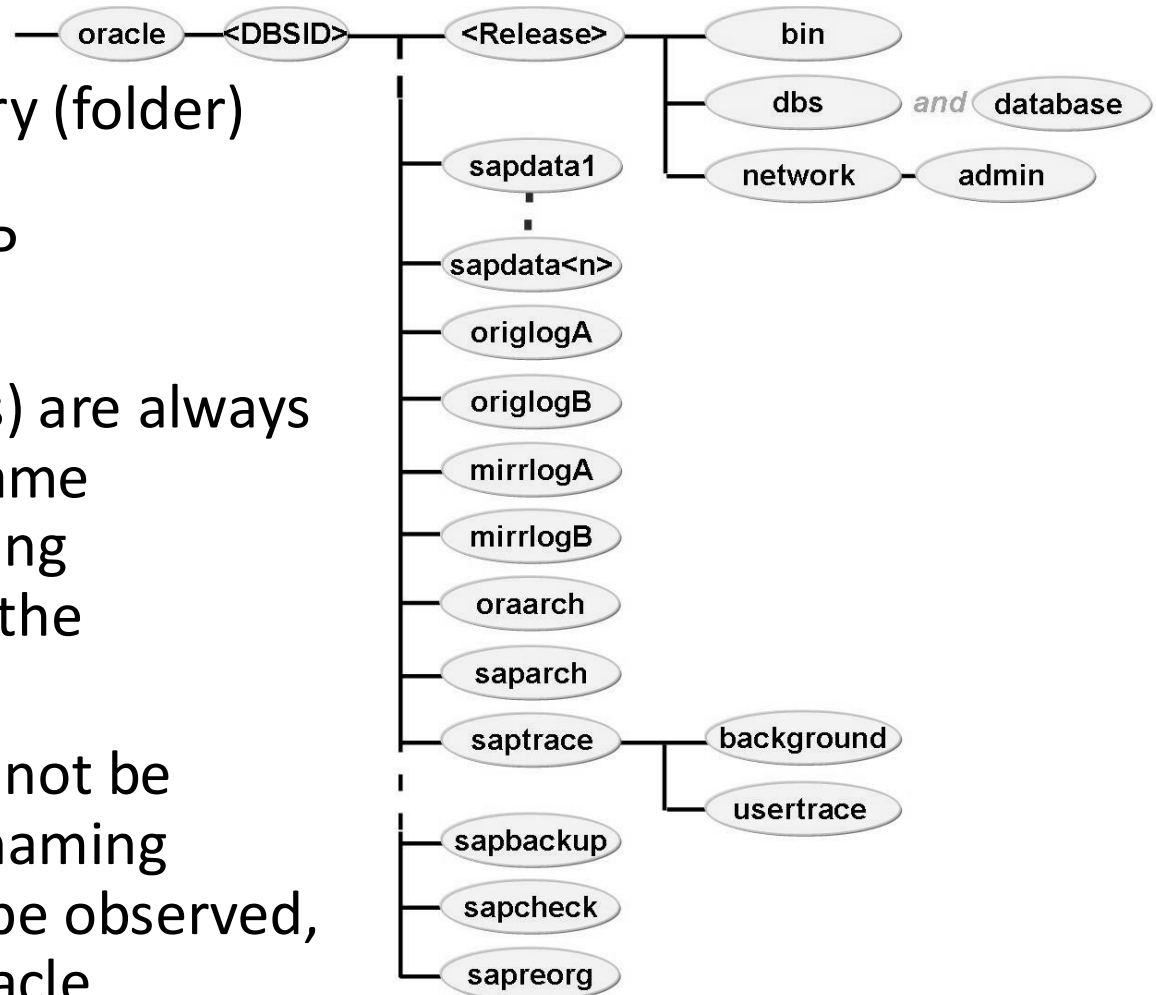
- `set <env_variable> <value>`
- sets an environment variable to requested value

Displays value of ORACLE_HOME environment variable

- `echo $ORACLE_HOME`

Overview of Database Oracle Directory Structure in SAP

- The Oracle directory (folder) and file names are standardized in SAP environments.
- Directories (folders) are always created with the same structure and naming convention during the installation.
- This structure may not be changed, and the naming conventions must be observed, as SAP tools for Oracle administration rely on it.



Overview of Database Oracle Directory Structure in SAP

db

- The Oracle profile init<DBSID>.ora or spfile<DBSID>.ora holds the Oracle instance configuration parameters
- The profile init<DBSID>.sap holds configuration parameters for administration tools BR*Tools

sapdata<n>

- Contains the data files of the tablespaces

Overview of Database Oracle Directory Structure in SAP

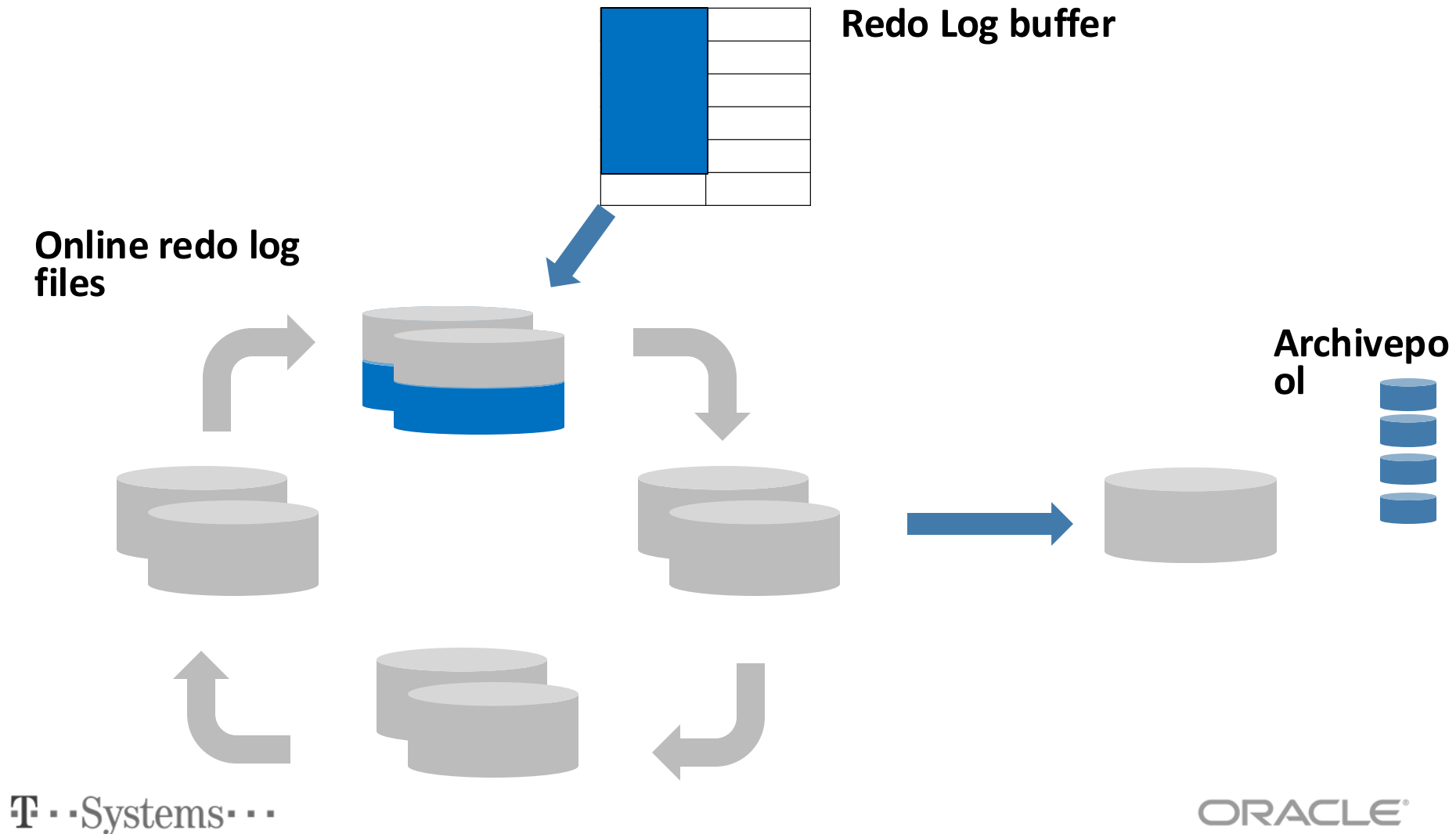
origlogA/B, mirrlogA/B

- Online redo log files reside in the origlog and mirrlog directories: Log file numbers 1 and 3 and their mirrors in origlogA and mirrlogA, log file numbers 2 and 4 and their mirrors in origlogB and mirrlogB, respectively

oraarch

- Offline redo log files are written to the oraarch directory; their names are specified with help of Oracle instance configuration parameters, so the name <DBSID>arch1_<LSN>.dbf is just an example.

Overview of Database Logging of Modifications



Overview of Database Oracle Directory Structure in SAP

saptrace

- Oracle dump files are written in the directory saptrace. The Oracle alert log alert_<DBSID>.log occurs in the directory background. Traces of Oracle shadow processes are written to the directory usertrace.

saparch

- Stores the logs written by the SAP tool BRARCHIVE

sapbackup

- Stores logs written by the SAP tools BRBACKUP, BRRESTORE, and BRRECOVER

Overview of Database Oracle Directory Structure in SAP

sapreorg

- BRSPACE creates logs for its different functions here

sapcheck

- BRSPACE creates logs for its different functions here

Overview of Database Oracle Directory Structure in SAP

Directory	Contents	File name examples
<Release>		
bin	Oracle executables	
db or database	SAP and Oracle profiles	init<DBSID>.ora, spfile<DBSID>.ora, init<DBSID>.sap
Network admin	Listener or client configuration files	listener.ora, tnsnames.ora
sapdata1 : : sapdata<n>	Data files	<DBSID>.DATA1, SYSTEM.DATA1 ROLL.DATA1, ctrl<DBSID>.dbf
origlogA	Online redo log files	log_g11m1.dbf, log_g13m1.dbf
origlogB	Online redo log files	log_g12m1.dbf, log_g14m1.dbf
mirrlogA	Online redo log files	log_g11m2.dbf, log_g13m2.dbf
mirrlogB	Online redo log files	log_g12m2.dbf, log_g14m2.dbf
oraarch	Offline redo log files	<DBSID>arch1_<LSN>.dbf
saparch	BRARCHIVE logs	arch<DBSID>.log
saptrace		
background	Oracle alert file	alert<DBSID>.log
usertrace	Trace files of server processes	<DBSID>_ora_<PID>.trc
sapbackup logs	BRBACKUP / BRRESTORE / BRRECOVER	
sapcheck	BRCONNECT logs	
sapreorg	BRSPACE logs, default compression directory	

Overview of Database Control file

Every Oracle database has a control file. A control file is a small binary file that records the physical structure of the database and includes:

- The database name
- Names and locations of associated datafiles and online redo log files
- The timestamp of the database creation
- The current log sequence number
- Checkpoint information

Location of the control file is configured in Oracle profile

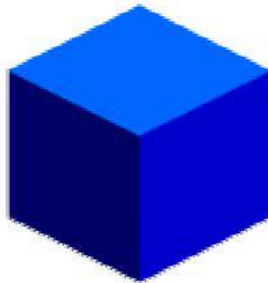
`init<SID>.ora`

Overview of Database

Logical and Physical Database Structures

Logical structure

- Database
- Tablespace
- Segment – Table / Index
- Extent
- Data block



Segment



Extents



Data
blocks



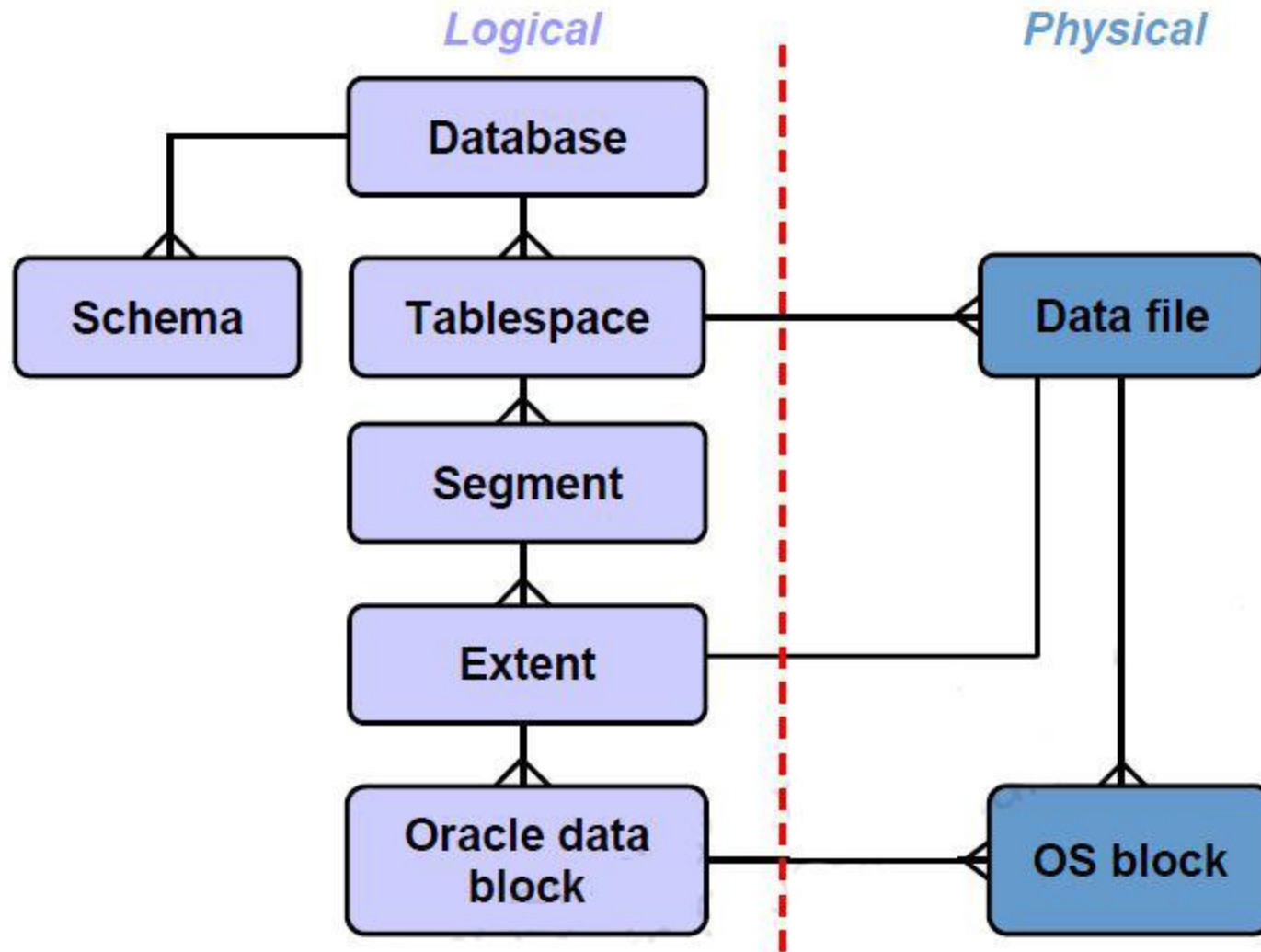
Disk
blocks

Physical structure

- Datafile
- Disk block

Overview of Database

Logical and Physical Database Structures



Database Administration Tools

Oracle Administration Tools on OS level

- BRTOOLS
- SQLPLUS
- BRGUI
- Enterprise Manager
- SQL Developer

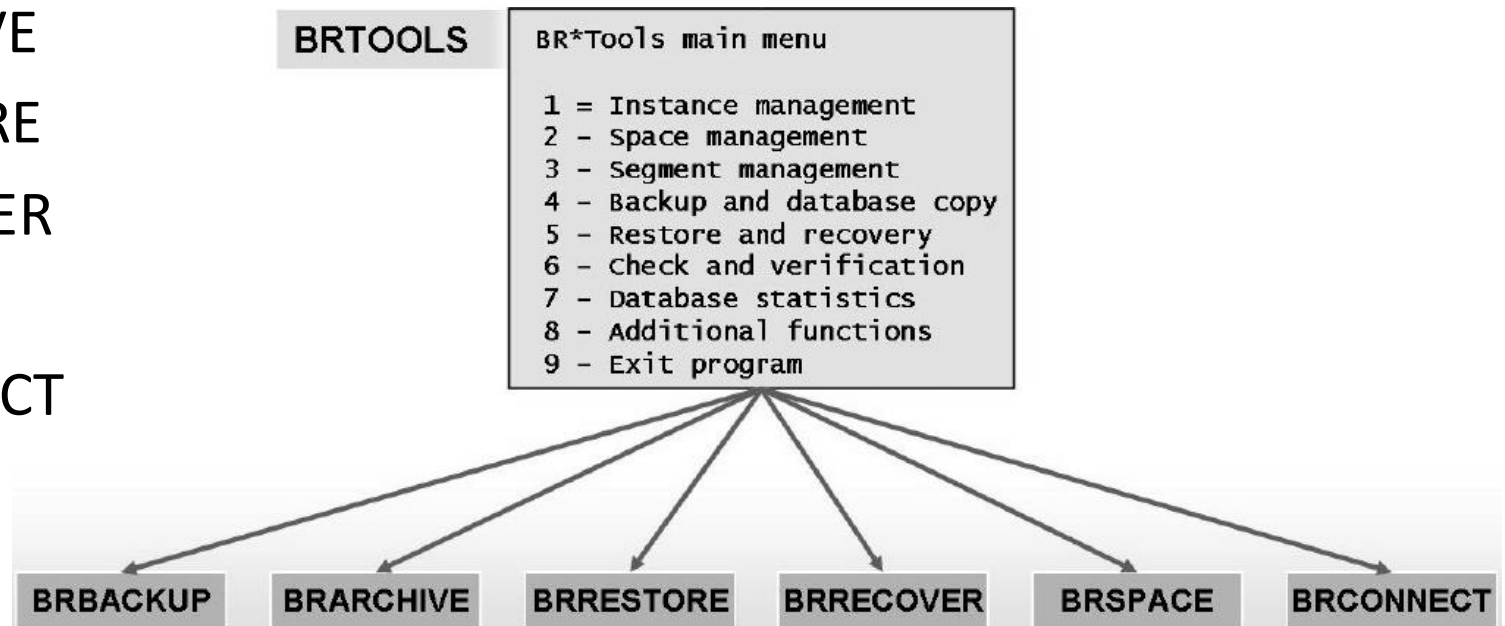


Database Administration Tools

BRTOOLS

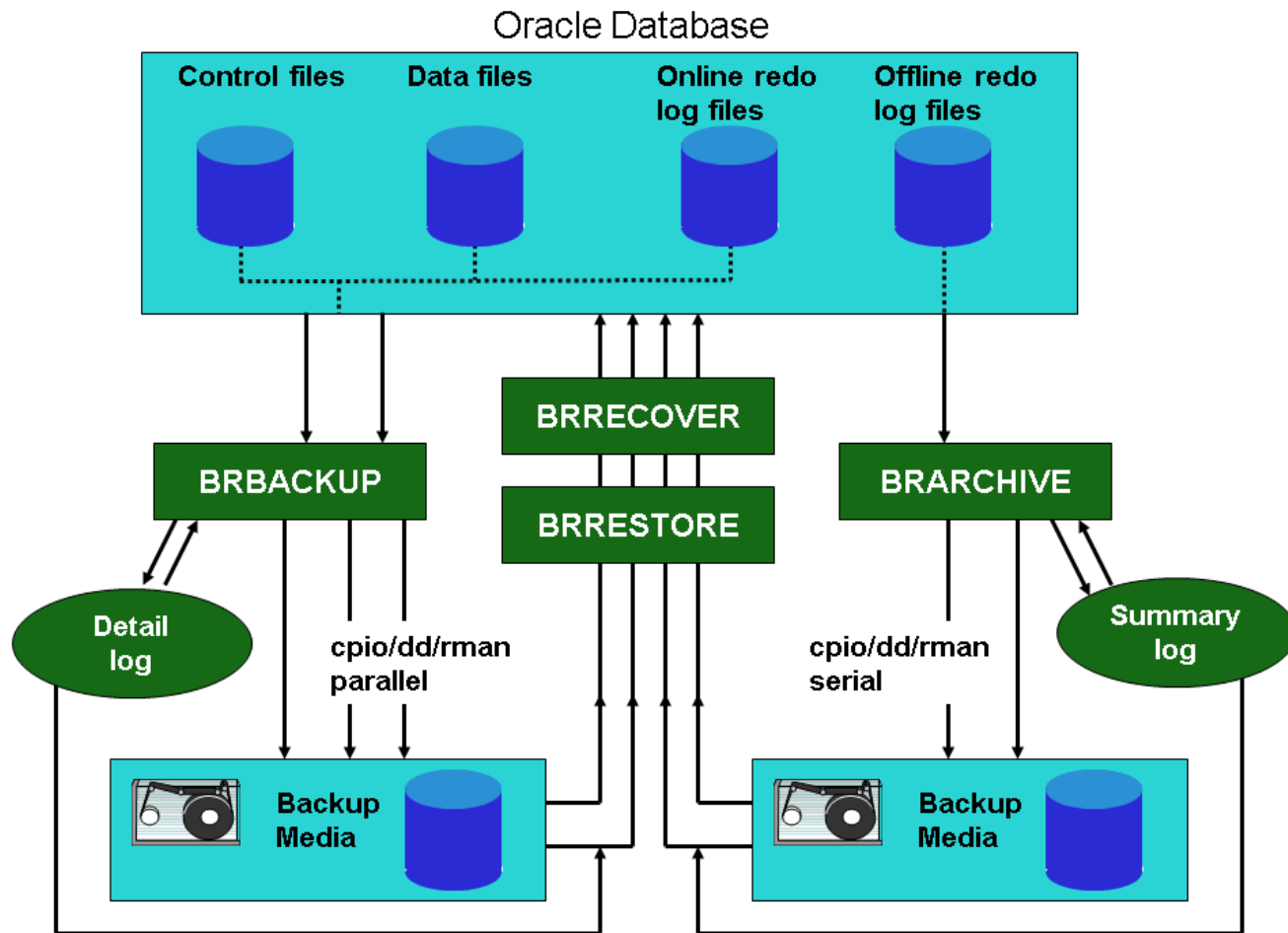
BRTOOLS is provided as a character-based user interface for the functional programs:

- BRBACKUP
- BRARCHIVE
- BRRESTORE
- BRRECOVER
- BRSPACE
- BRCONNECT



Database Administration Tools

BRTOOLS - Menu Structure



Space Management

Tablespaces

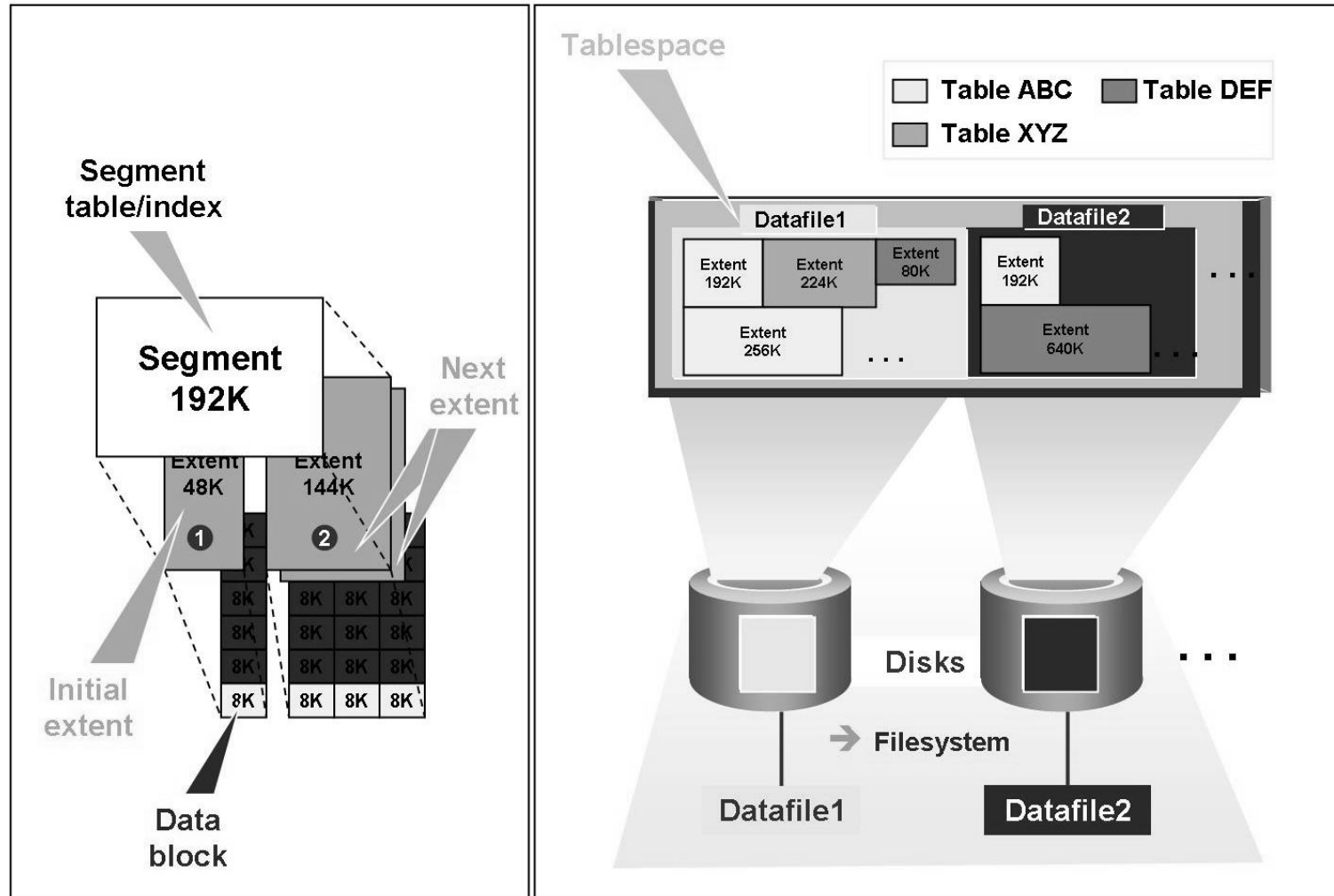
Oracle stores data logically in tablespaces and physically in datafiles associated with the corresponding tablespace

Possible actions on tablespace

- Resize Extend / Shrink
- Add
- Change autoextend properties
- Set offline
- Drop

Space Management

Tablespaces Storage Parameters

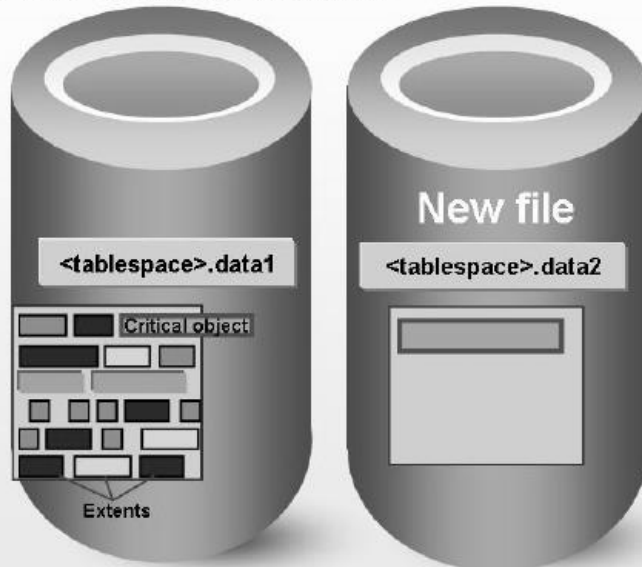


Space Management

Enlarging a Tablespaces

ADD A DATA FILE:

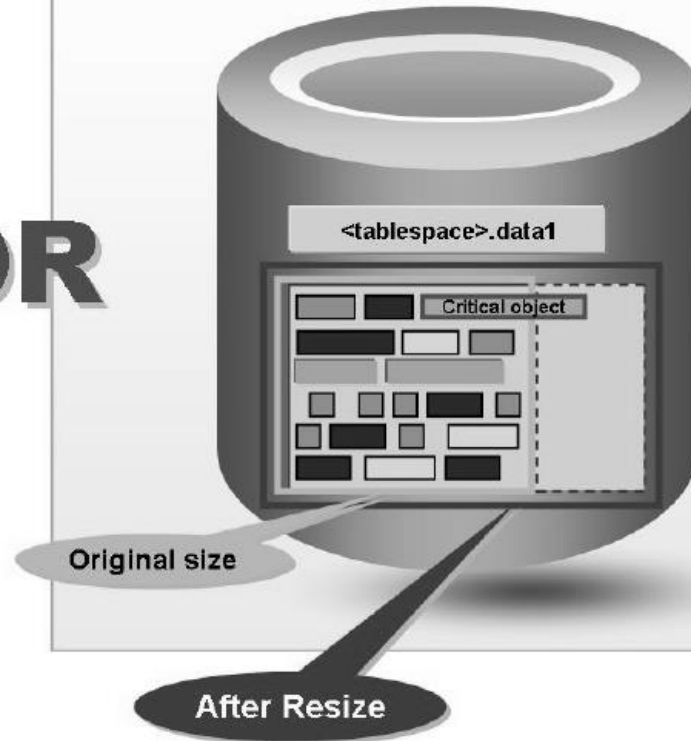
File size depends on the estimated increase of the tablespace objects. Check for the number of data files in the database.



OR

RESIZE THE DATA FILE:

Extend the size of the data file depending on the space available on the file system and size of critical object.



Space Management

Tablespaces - Naming Conventions

Tablespace Name	Segment Type	Contents
SYSTEM	tables, indexes, rollback	Oracle Data Dictionary
SYSAUX	tables, indexes	mandatory help tablespace for the SYSTEM tablespace
PSAPROLL or PSAPUNDO	rollback/undo	only rollback/undo segments
PSAPTEMP	temp. segments	only temporary segments
PSAP<SCHEMA-ID>	tables,	objects of the SAP Web AS
PSAP<SCHEMA-ID>USR	tables, indexes	customer objects
PSAP<SCHEMA-ID><REL>	tables, indexes	release-dependent data (ABAPs, loads)
PSAP<SCHEMA-ID>DB	tables, indexes	objects of the SAP Web AS Java schema (starting with SAP Web AS 6.40)

Administering Oracle Instances

Statuses of an Oracle Instance / Startup options



OFFLINE

Oracle instance processes and buffers do not exist

NOMOUNT

Read parameter file, start processes, allocate memory buffers

MOUNT

Open control files

OPEN

Open all data files and online redo log files

Administering Oracle Instances

Statuses of an Oracle Instance - NOMOUNT

NOMOUNT

- In the NOMOUNT phase, the parameter file is opened and evaluated, and the database instance is started. Operating system resources are allocated using configuration information stored in the parameter file.

Administering Oracle Instances

Statuses of an Oracle Instance - MOUNT

MOUNT

- In the MOUNT phase, the control files of the database are opened (using the value of parameter `CONTROL_FILES` from the parameter file) and evaluated. The system reads the information about the physical structure of the database. Although data files and log files are not yet opened, parts of the data dictionary are loaded, so V\$ views are available. If one or more of the files specified by the `CONTROL_FILES` initialization parameter does not exist or cannot be opened when you attempt to mount a database, Oracle returns an error message and does not mount the database.

Administering Oracle Instances

Statuses of an Oracle Instance - OPEN

OPEN

- In the OPEN phase, all remaining files of the database system are opened. If required, an instance recovery is performed during opening the database. Only when the database is open can valid database users connect to the database. If one or more of the data files or redo log files is not available or cannot be opened when attempting to open a database, Oracle returns an error message and does not open the database.

Administering Oracle Instances

Statuses of an Oracle Instance / Startup options

- The **NOMOUNT** state is necessary for creating a database and for re-creating lost control files.
- The **MOUNT** state is needed for database recovery, for changing the ARCHIVELOG mode, for renaming (moving) data files, and for adding, dropping, or renaming online redo log files.

Administering Oracle Instances

Statuses of an Oracle Instance / Shutdown options

NORMAL

- No new connections are allowed. Oracle waits for all currently connected users to disconnect from the database. Only after the last user has disconnected (in SAP system: all work processes have been stopped), Oracle shuts down the database: all files are closed, the database is dismounted, and the instance is shut down.

TRANSACTIONAL

- After this instruction has been defined, no new connections or new transactions are allowed. Oracle waits for all open transactions to finish, then it disconnects all database connections and shuts down the database.

Administering Oracle Instances

Statuses of an Oracle Instance / Shutdown options

IMMEDIATE

- No new connections are allowed, and no new transactions are allowed to be started after the statement is issued. The PMON process ends all user sessions and performs a rollback of any open transactions. Then the database is shut down.

ABORT

- No new connections are allowed, and no new transactions are allowed to be started after the statement is issued. All client SQL statements currently being processed are terminated, without rolling back open transactions. User connections are disconnected and Oracle processes are stopped.

Administering Oracle Instances

Statuses of an Oracle Instance / Shutdown options

On the way down:

- Uncommitted changes rolled back, for **IMMEDIATE**
- Database buffer cache written to data files
- Resources released

During

SHUTDOWN

NORMAL

or

SHUTDOWN

TRANSACTIONAL

or

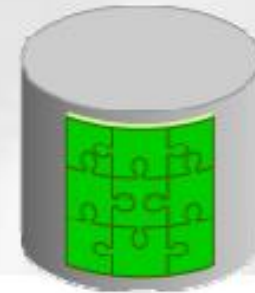
SHUTDOWN

IMMEDIATE

On the way up:

- No instance recovery

Consistent database
(clean database)



Administering Oracle Instances

Statuses of an Oracle Instance / Shutdown options

On the way down:

- Modified buffers not written to data files
- Uncommitted changes not rolled back



During
SHUTDOWN ABORT
or
Instance failure

Inconsistent database
(dirty database)

On the way up:

- Online redo log files used to reapply changes
- Undo segments used to roll back uncommitted changes
- Resources released

Introduction to SQL SQLplus

With the Oracle tool SQL*Plus, you can:

- Start and stop the database
- Log into the database
- Perform database administration
- Enter SQL*Plus commands to configure an SQL*Plus environment
- Enter, edit, store, retrieve, and run SQL commands and SQL scripts
- Redirect the output of query results to text files

Introduction to SQL

SQL commands

SQL commands by type

- DML – SELECT, INSERT, UPDATE, DELETE, MERGE
- DDL – CREATE, ALTER, DROP, RENAME, TRUNCATE, COMMENT
- DCL – GRANT, REVOKE
- TCL – COMMIT, ROLLBACK, SAVEPOINT

Introduction to SQL

SQL commands

sqlplus / as sysdba

Basic syntax

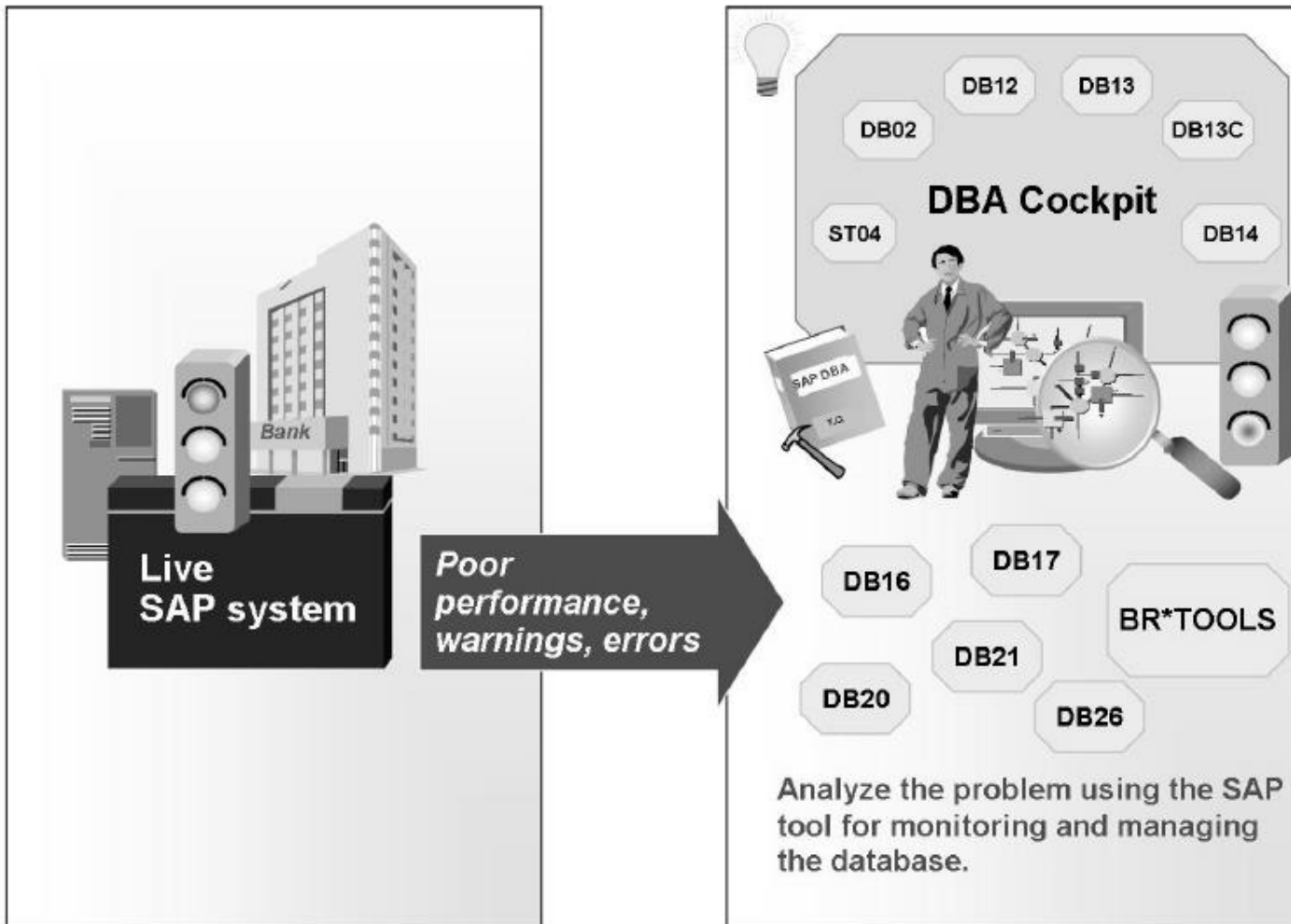
SELECT <COLUMN> from <SCHEMA>.<TABLE> where
<COLUMN>='<condition>';

UPDATE <SCHEMA>.<TABLE> set <COLUMN>=<value> where
<COLUMN>='<condition>';

DELETE from <SCHEMA>.<TABLE> where <COLUMN>='<condition>';

SAP DB Monitors basics

DB12, DB13, DB02...



SAP DB Monitors basics

DB13

The **DBA Planning Calendar** and the central DBA planning calendar (transaction DB13 and DB13C) is available for scheduling backups and other

administrative jobs in your database system. You can use this transaction to schedule backups and administrative activities locally or centrally for several SAP systems and databases.

SAP DB Monitors basics

DB12, DB14

You can use the **Backup Log Overview** (transaction DB12) to display the results of your backups and the status of the archive directory. If all backups are available for a restore and recovery, it also contains a function for checking the restore reports.

The **DBA Operations Monitor** (transaction DB14) checks the status and logs of all database operations, including backup monitoring, updates of the optimizer statistics, and database checks.

SAP DB Monitors basics

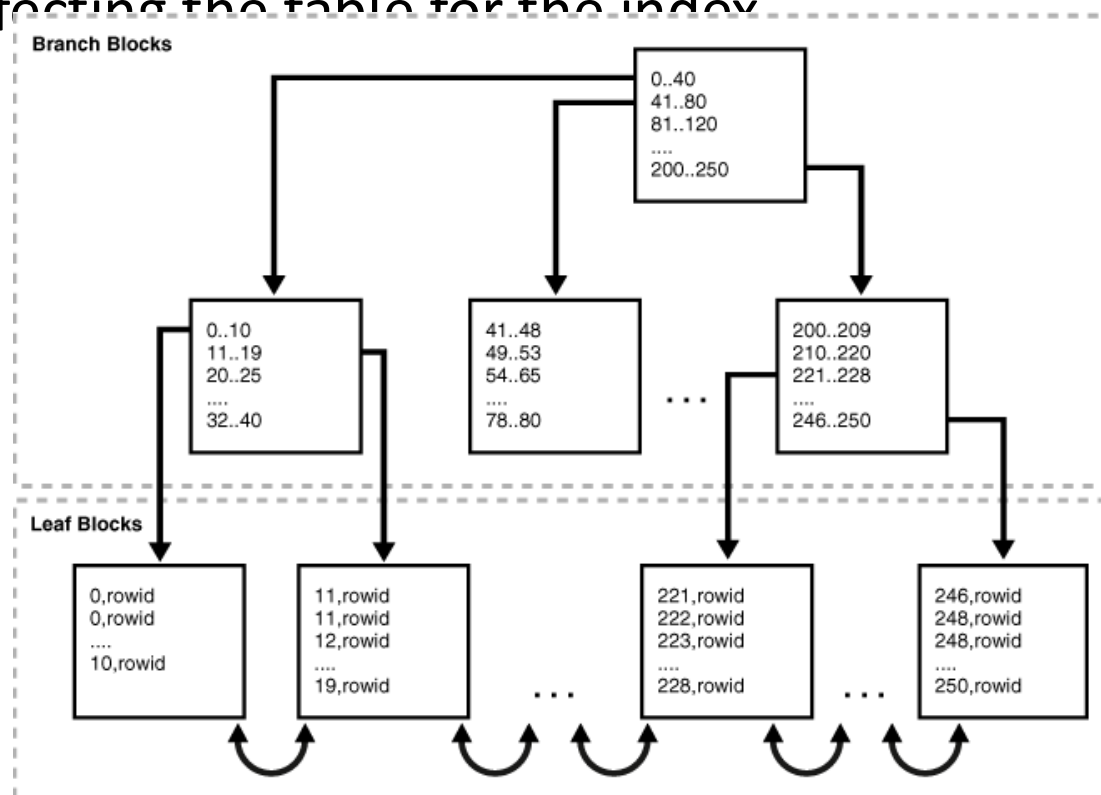
ST04, DB02

The **Database Performance Monitor** (transaction ST04) displays the most important indicators for Oracle database performance – such as buffer cache quality, statistics of user calls or number of block reads per SQL statement – and the configuration of the database management system and the database, either directly through showing current parameter values or with help of V\$ views.

- The **Tables and Indexes Monitor** (transaction DB02) monitors the storage behavior of the database (for example, space statistics showing the history of the database, or size and free space in each tablespace) and the status of the database objects (for example size of each table in kilobytes and blocks, or indexes that are defined in the ABAP Dictionary but missing in the database).

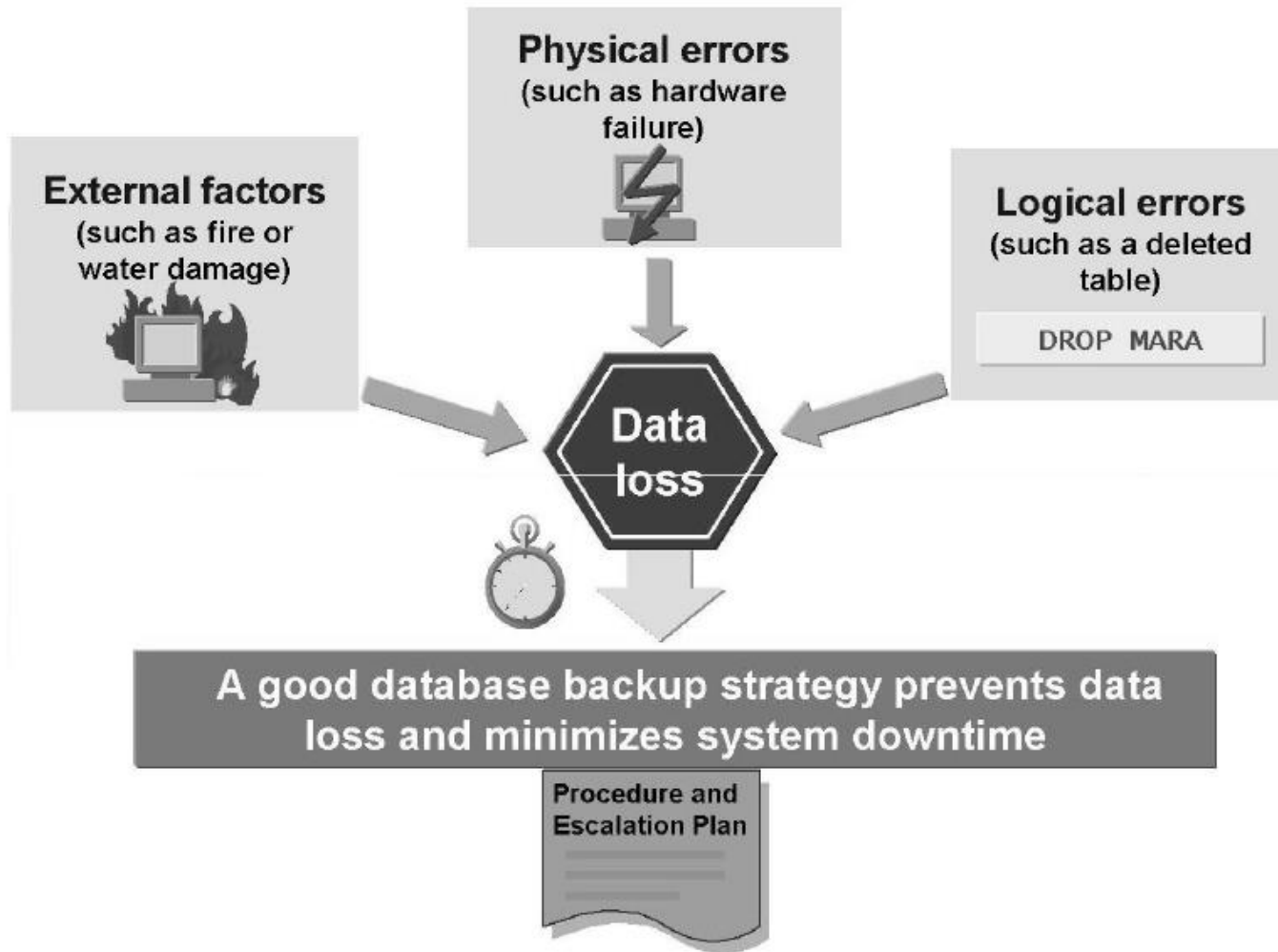
Database index

Indexes are schema objects that are logically and physically independent of the data in the objects with which they are associated. Thus, an index can be dropped or created without physically affecting the table for the index.



Administering Oracle Instances

DB Backup

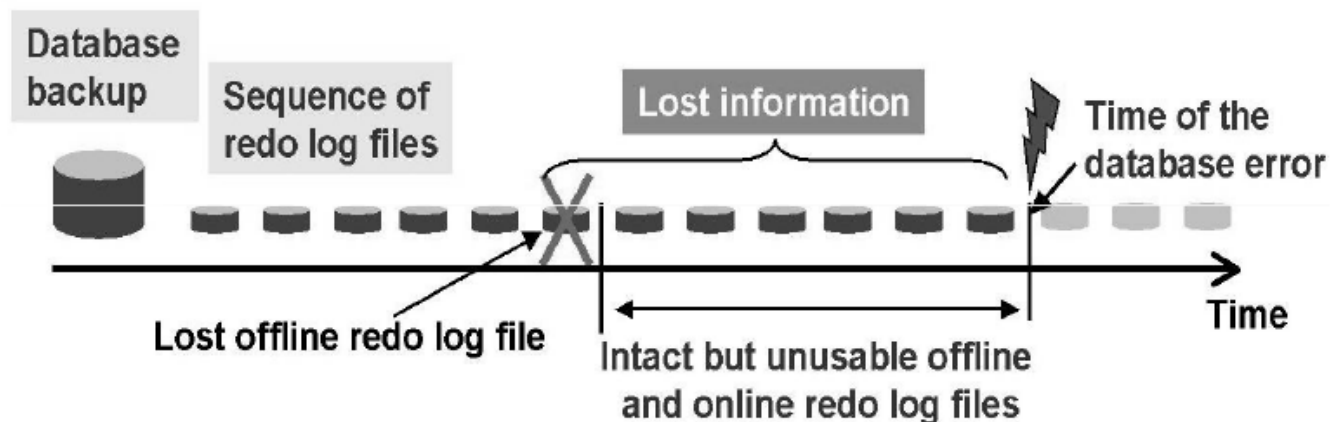


Administering Oracle Instances

DB Backup

Forward recovery

A database backup is restored and
you want to recover data from redo log files

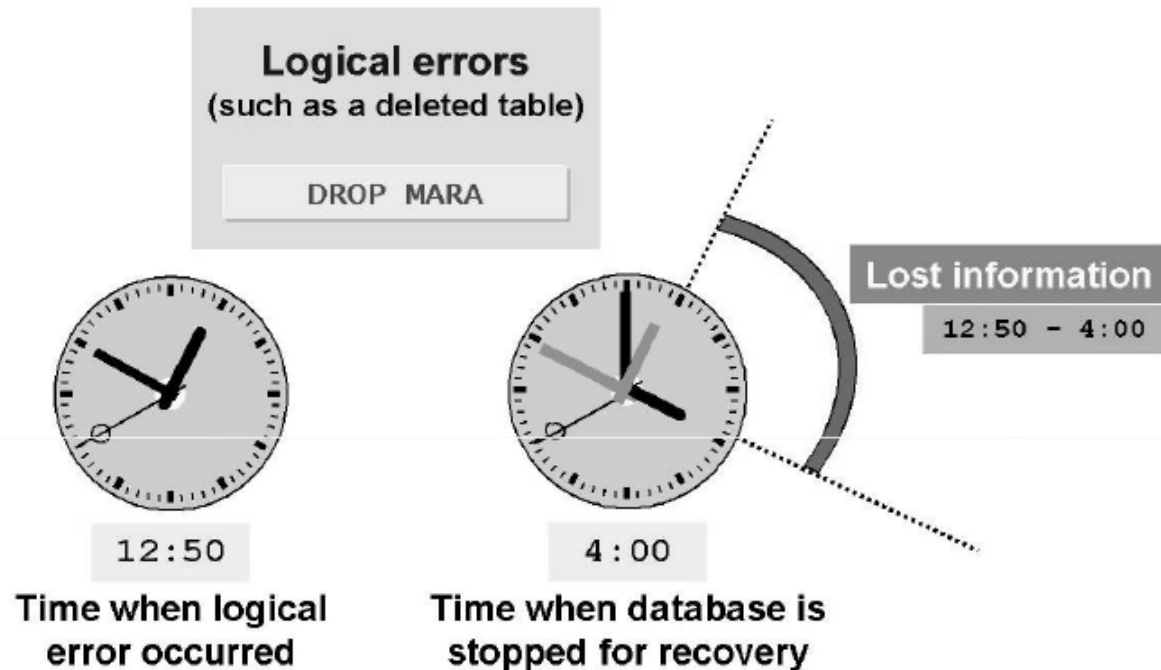


**If one offline redo log file is lost,
none of the files that follow it can be used**

Administering Oracle Instances

DB Backup

Point-in-time recovery



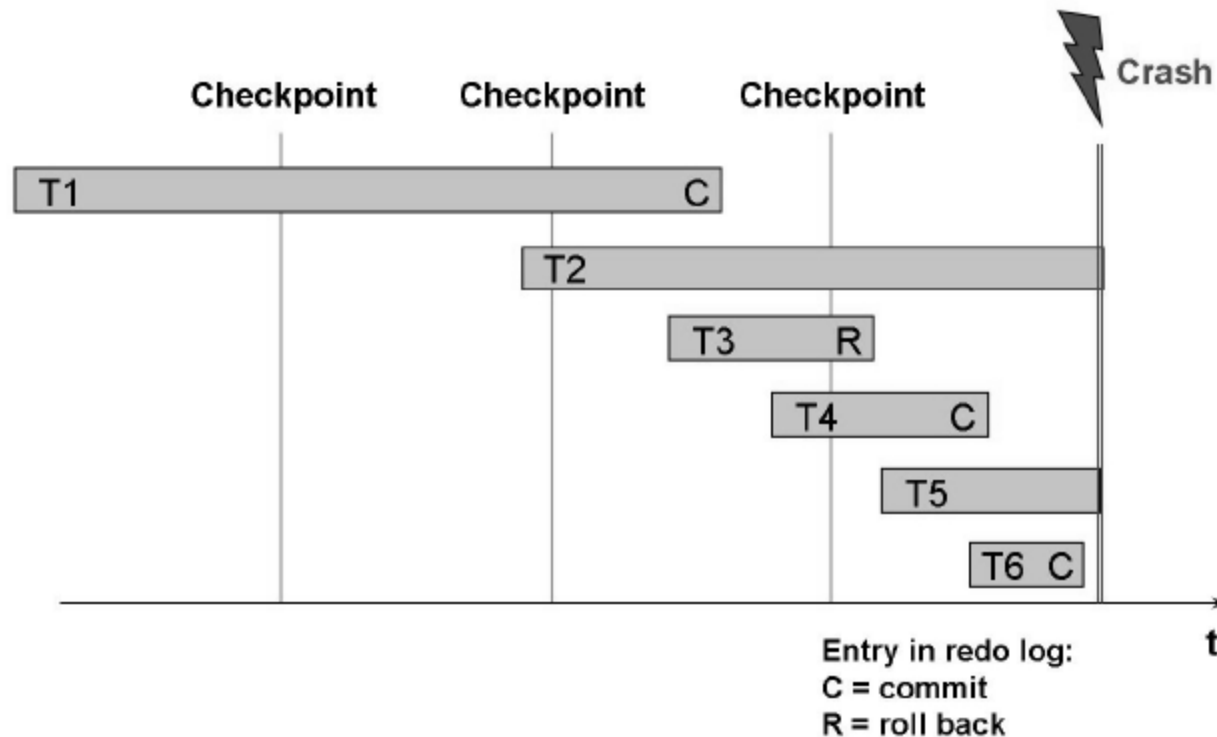
You must recover the complete database to a point in time before the error

Point-in-time recovery is very critical in a system landscape with data dependencies between systems

Administering Oracle Instances

DB Recovery

Database Recovery



Roll forward (read redo log): T2, T3, T4, T5, T6
Roll back (read undo space): T2, T3, T5
No recovery : T1

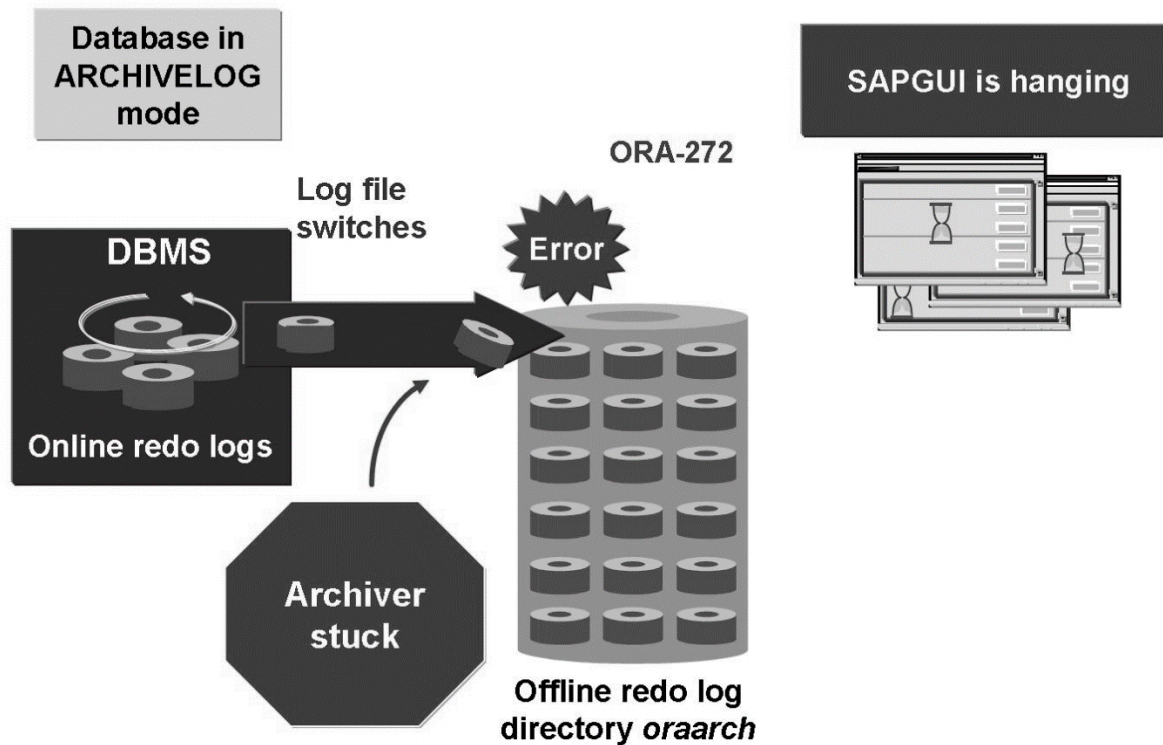
Troubleshooting

Log files

- Backup log
 - /oracle/<SID>/sapbackup/*.afd
- Redologs backup log
 - /oracle/<SID>/saparch/arch<SID>.log
- Alert log
 - 11g / 12c
 - /oracle/<SID>/saptrace/diag/rdbms/<sid>/<SID>/trace/alert_<SID>.log
 - 10g
 - /oracle/<SID>/saptrace/background/alert_<SID>.log

Troubleshooting Archiver Stuck

An archiver stuck is a situation in which the archiver instance process cannot copy an online redo log file to the archive directory (oraarch) and the online redo log file to be copied needs to be overwritten



Oracle DBA For SAP

